■ PHASE 0 — Prerequisite Mindset (Skip if Already Strong)
■ Goal

You should be comfortable writing clean structured code, not just solving DSA problems.

■ Must Know Before LLD

Functions

Classes

Objects

Basic STL / Collections

Clean coding habits

■ Practice (Warmup)

Implement Stack using Class

Implement Queue using Class

Implement Bank Account class

Implement Student Management mini system

■ Resources

■ YouTube:

CodeWithHarry – OOP in C++

Apna College – OOP Complete Playlist

■ PHASE 1 — OOP MASTER LEVEL (VERY IMPORTANT)

■ Many students rush LLD → Big mistake
■ LLD = Applied OOP

■ 1■■ OOP Core Concepts (Deep Level)
■ Class & Object

Subtopics:

Constructors (Default, Param, Copy)

Destructor

Static variables

Static methods

Object lifecycle

■ Encapsulation
Private / Public / Protected

Getters Setters

Data hiding

Real world mapping

■ Abstraction

Abstract class

Interface

Pure virtual functions

When to use interface vs abstract class

■ Inheritance (Deep)

Types:

Single

Multilevel

Multiple

Hybrid

Hierarchical

Concepts:

Method overriding

Virtual functions

Diamond problem

■ Polymorphism

Compile time (Overloading)

Runtime (Overriding)

Operator overloading (Optional but good)

■ Practice Questions (MANDATORY)

Implement:
■ Library Management System
■ Hotel Booking Basic Model
■ Online Shopping Basic Model
■ Employee Payroll System

■ Best YouTube Playlists

■ Neso Academy – OOP

■ Concepts with Code – OOP LLD Intro
■ Gaurav Sen – OOP Basics for Design

■ PHASE 2 — SOLID PRINCIPLES (INTERVIEW FAVORITE)
■ 2■■ SOLID Principles (Extreme Depth)
■ S — Single Responsibility Principle

One class → One job

Practice:

Split God class into smaller classes

■ O — Open Closed Principle

Open for extension
Closed for modification

Practice:

Payment system → Add new payment without editing old code

■ L — Liskov Substitution

Child should behave like parent logically

■ I — Interface Segregation

Small interfaces > Big interfaces

■ D — Dependency Inversion

Depend on abstraction not concrete class

■ Practice Design Tasks

■ Payment Gateway with multiple payment types
■ Notification system (Email + SMS + Push)
■ Logging framework

■ Resources

■ Gaurav Sen – SOLID Principles
■ Concepts with Code – SOLID Playlist

■ PHASE 3 — DESIGN PATTERNS (LLD CORE WEAPON)
■ 3■■ Creational Patterns

Singleton

Factory

Abstract Factory

Builder

Practice:

Logger Singleton

Vehicle Factory

Database Connection Singleton

■ 4■■ Structural Patterns

Adapter

Decorator

Facade

Proxy

Practice:

Payment adapter

Coffee decorator pricing

■ 5■■ Behavioral Patterns

Strategy

Observer

State

Command

Practice:

Notification Observer

Payment Strategy

Order State Machine

■ Resources

■ Refactoring Guru (Website + YouTube)
■ Gaurav Sen – Design Patterns
■ Concepts with Code – Design Patterns

■■ PHASE 4 — CORE LLD INTERVIEW PROBLEMS
■ Beginner LLD Problems

Solve FULL class diagram + code

■ Parking Lot
■ Library System
■ Movie Ticket Booking
■ ATM Machine
■ Car Rental System
■ Intermediate LLD Problems

■ BookMyShow
■ Food Delivery (Swiggy type)
■ Ride Sharing (Uber basic)
■ Splitwise
■ Chess Game

■ Advanced LLD Problems

■ Rate Limiter
■ Notification System (Scalable)
■ Cache System (LRU / LFU class design)
■ Messaging Queue Basic Model

■ Resources

■ Gaurav Sen – LLD Problems
■ Concepts with Code – LLD Interview Series
■ Anuj Bhaiya – LLD Playlist

■ PHASE 5 — UML & DESIGN COMMUNICATION
■ Learn

Class Diagram

Sequence Diagram

Use Case Diagram

■ Resources

■ Lucidchart UML Tutorials
■ Visual Paradigm UML Basics

■ PHASE 6 — DATABASE + API THINKING (LLD+)
Learn Basics

Entity design

Normalization

REST API basics

DTO concept

Practice

Design:

User Service API

Order Service API

Payment Service API

■ PHASE 7 — MOCK INTERVIEW LEVEL
Full Case Study Practice
Do end-to-end:

■ Parking Lot → Diagram → Code → Edge cases
■ Uber → Modules → Classes → Flow
■ Splitwise → Ledger logic → Settlement

■ IDEAL TIMELINE (REALISTIC)
■ Month 1

OOP + Practice Systems

■ Month 2

SOLID + Design Patterns

■ Month 3

LLD Problems + Mock Interviews

■ TOP PLAYLIST COMBO (If You Follow Only Few)

If limited time:

■ MUST WATCH

Concepts with Code – LLD Playlist

Gaurav Sen – LLD + System Design Basics

Refactoring Guru – Design Patterns

■ EXTREME BONUS (If Targeting Top Product Companies)

Learn:

Thread Safety Basics

Immutability

Basic Concurrency

Caching concepts

■ FINAL INTERVIEW READY CHECKLIST

You are LLD ready if you can:

■ Convert problem → Classes
■ Apply SOLID naturally
■ Use 2–3 design patterns naturally
■ Write clean extensible code
■ Explain tradeoffs